

Continuous Control Assessment

Sentinel Nexus — proven control posture (generated from the test run)

Information Security & AI Governance

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Assessment Summary

Unlike the *claims* in the control catalog, every status below is **derived from the pipeline’s JUnit results in this run** — a control is Satisfied only when its bound tests actually executed and passed.

Assessed **36** controls — 30 Satisfied · 0 Failed · 4 Not-Run · 2 Documentation.

Status	Count
Satisfied	30
Failed	0
Not-Run	4
Documentation	2

Framework Posture

Coverage = the share of **applicable** framework items addressed by at least one **Satisfied** control (claimed *and* proven).

Framework	Covered	Applicable	Posture
OWASP Top 10 LLM	9	10	90.0%
MITRE ATLAS	8	8	100.0%
NIST AI 600-1	13	14	92.9%
NIST SP 800-53	28	34	82.4%
NIST CSF 2.0	6	6	100.0%

Per-Control Proven Status

ID	Title	Intent	Assessed	Evidence	Bound tests	Code evidence
AI-GROUNDING	Confabulated-evidence grounding	implemented	Satisfied	complete	TestGroundingEnforcement	analytics/llm_hunter/agent/L74
AI-MEMORY-TTL	Immunity-memory TTL / expiry	implemented	Satisfied	complete	TestMemoryTTL	analytics/llm_hunter/agent/L175
AI-PROVENANCE	AI-origin provenance disclosure	implemented	Satisfied	complete	TestProvenanceDisclosure	analytics/llm_hunter/agent/L192
AI-REVIEW-BOARD	Adversarial review board (per-expert counterparts)	implemented	Satisfied	complete	test_review_board_simulation.py	analytics/llm_hunter/orchestrator/L140
IAC-HARDENING	OS / kernel / network hardening baseline	implemented	Not-Run	complete	test_infrastructure_hardenig/run	hardening/tasks/main.yml/L22
ING-DLQ-BREAKER	Durable worker circuit breaker + dead-letter routing	implemented	Satisfied	complete	TestWorkerS3ArchiveDLQ TestCognitive-FaultDLQ	tools/lib_siem_core/src/L60
ING-ZERO-TRUST	Zero-Trust ingestion gateway (HMAC + 3-tier replay defense)	implemented	Satisfied	complete	TestEvidenceIngress	services/core_ingress/src/L213
NC-1-BIAS-AUDIT	Bias/disparity + homogenization scheduled audit	implemented	Satisfied	complete	TestBiasAudit	analytics/llm_hunter/agent/L249
NC-10-VERDICT-LINEAGE	Tamper-evident verdict lineage	implemented	Satisfied	complete	TestVerdictLineage TestVerdictLedger	analytics/llm_hunter/agent/L180
NC-11-ENERGY-ACCOUNTING	Per-run inference energy accounting	implemented	Satisfied	complete	TestInferenceEnergy TestEnergyAccounting	analytics/llm_hunter/agent/L189
NC-2-CALIBRATION	Confidence-calibration ledger	implemented	Satisfied	complete	TestCalibrationLedger	analytics/llm_hunter/agent/L133

ID	Title	Intent	Assessed	Evidence	Bound tests	Code evidence
NC-3-FRONTIER-PIN	Frontier model boot-time version-pin enforcement	implemented	Satisfied	complete	TestFrontierPinEnforcement	analytics/llm_hunter/agent/llm_hunter/L201
NC-4-RETENTION	Data retention & decommissioning policy	documented	Documentation	complete	(none)	—
NC-6-ENERGY	Environmental impact estimate	documented	Documentation	complete	(none)	—
NC-7-ENDPOINT-ABUSE	Inference-endpoint abuse / model-extraction monitoring	implemented	Satisfied	complete	TestEndpointQuota, TestVolumeAnomaly, TestMembershipInferenceSignal, TestEndpointAbuseReport, TestEndpointAbuseMonitor	analytics/llm_hunter/agent/llm_hunter/L531
NC-8-OVER-RELIANCE	Automation-bias / over-reliance measurement	implemented	Satisfied	complete	TestOverReliance, TestRelianceLedger	analytics/llm_hunter/agent/llm_hunter/L365
NC-9-ACTIVE-LEARNING	Active-learning failure capture	implemented	Satisfied	complete	TestActiveLearningFailureCapture, TestActiveLearning	analytics/llm_hunter/agent/llm_hunter/L201
SEC-BLAST-RADIUS	Blast-radius cap & entity state machine	implemented	Satisfied	complete	test_hunter_contract_state	analytics/llm_hunter/state/L290
SEC-CANARY	Canary token prompt-leak tripwire	implemented	Satisfied	complete	test_agentic_swarm_contract_state	analytics/llm_hunter/orchestrator/L233
SEC-DLP-EGRESS	Outbound DLP / sovereign data isolation	implemented	Not-Run	complete	test_cognitive_bypass_tripwire	analytics/llm_hunter/tools/L73
SEC-DUCKDB-SANDBOX	Read-only data-lake query sandbox	implemented	Satisfied	complete	test_query_cookbook	analytics/llm_hunter/tools/L30

ID	Title	Intent	Assessed	Evidence	Bound tests	Code evidence
SEC-ENDPOINT-ID	Endpoint identity injection defense	implemented	Satisfied	INCOMPLETE — logic-only (no wired invocation→execution)	TestEndpointIdInjectionDefense	llm_defense_core/src/mocks/defense/endpoint_id_injection_defense.py L20
SEC-FAILOVER	Cascading LLM failover & sovereign degradation	implemented	Satisfied	complete	TestLLMCircuitBreaker	analytics/llm_hunter/agent/llm_circuit_breaker.py L25
SEC-IDEMPOTENT-SOAR	Idempotent SOAR execution & deduplication	implemented	Satisfied	complete	test_agentic_swarm_orchestrator	analytics/llm_hunter/agent/agentic_swarm_orchestrator.py L377
SEC-MODEL-DOS	Model denial-of-service bounding	implemented	Satisfied	complete	test_agentic_swarm_orchestrator	analytics/llm_hunter/agent/agentic_swarm_orchestrator.py L56
SEC-OUTPUT-SCHEMA	Strict SOAR output-contract enforcement	implemented	Satisfied	complete	test_agentic_swarm_orchestrator	analytics/llm_hunter/agent/agentic_swarm_orchestrator.py L174
SEC-REGRESSION-GATE	Deterministic regression / deploy gate	implemented	Satisfied	complete	test_mlops_serving_mlops	mlops/scripts/03_eval_model_regression_gate.py L338
SEC-RLHF-QUARANTINE	Sybil RLHF poisoning quarantine	implemented	Satisfied	complete	TestNATSSubjectAutoscaler	services/worker_rlhf/src/worker_rlhf.py L91
SEC-SANITIZER	Cognitive boundary isolation & untrusted-payload wrapping	implemented	Not-Run	complete	test_cognitive_bypassing	analytics/llm_hunter/tools/cognitive_bypassing.py L48
SEC-SUPPLY-CHAIN	Cryptographic model supply-chain integrity (SHA-384)	implemented	Satisfied	complete	test_mlops_serving_mlops	mlops/serve_vllm.sh:L48-L66
SEC-TRAINING-HYGIENE	Training-data hygiene & credential scrubbing	implemented	Satisfied	INCOMPLETE — logic-only (no wired invocation→execution)	test_mlops_serving_mlops	mlops/scripts/01_spool_data_hygiene.py L60
SEC-VECTOR-DIM	Vector dimensionality validation	implemented	Satisfied	complete	test_hunter_contract_validation	analytics/llm_hunter/tools/contract_validation.py L50

ID	Title	Intent	Assessed	Evidence	Bound tests	Code evidence
SIEM-CONFIG-CONTRACT	SIEM config ↔ fanout index contract	implemented	Satisfied	complete	test_siem_config.py	analytics/llm_hunter/agent/llm.py L140
SIEM-COUNTERPART-DISPROOF	Review-board counterpart SIEM disproof	implemented	Satisfied	complete	test_siem_review_board.py	analytics/llm_hunter/agent/llm.py L127
SIEM-E2E	SIEM federation end-to-end conservation	implemented	Not-Run	complete	test_siem_federation.py	test_siem_federation.py L195
SIEM-TOOL-GUARD	SIEM query tool — read-only / bounded / allowlist	implemented	Satisfied	complete	test_siem_query.py	analytics/llm_hunter/tool/llm.py L128

Open Findings (Plan-of-Action seed)

Controls claimed **implemented** with a proof gap: bound tests that **failed / did not run**, or an **incomplete evidence chain** (e.g. a lone logic snippet with no wired invocation→execution). A Failed or Incomplete finding turns the build red; a Not-Run finding is a coverage gap (blocking only in `--gate --strict / --full`).

- **IAC-HARDENING — OS / kernel / network hardening baseline** → *Not-Run*. Bound: `infrastructure/tests/test_infrastructure.py` [not-run]
- **SEC-DLP-EGRESS — Outbound DLP / sovereign data isolation** → *Not-Run*. Bound: `tests/lab_redteam/test_cognitive_bypass.py` [not-run]
- **SEC-ENDPOINT-ID — Endpoint identity injection defense** → *Satisfied*. Bound: `tests/test_worker_contracts.py::TestEndpointIdInjectionDef` [resolved=pass] · **evidence:** logic-only (no wired invocation→execution)
- **SEC-SANITIZER — Cognitive boundary isolation & untrusted-payload wrapping** → *Not-Run*. Bound: `tests/lab_redteam/test_cognitive_by` [not-run]
- **SEC-TRAINING-HYGIENE — Training-data hygiene & credential scrubbing** → *Satisfied*. Bound: `tests/lab_mlops_serving/test_mlops_serv` [resolved=pass] · **evidence:** logic-only (no wired invocation→execution)
- **SIEM-E2E — SIEM federation end-to-end conservation** → *Not-Run*. Bound: `tests/lab_siem_federation/test_siem_federation_e2e.py` [not-run]